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An AI-Driven Adaptive Cybersecurity Framework for IoT: Anomaly Detection, Dynamic Thresholding, Un-Contaminated Reinforcement Learning Defense, Post-Quantum Cryptography, and Tamper-Evident Audit Logging

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Open-Access Code Repository: https://github.com/Tanish-boop/AI-Driven-IoT-Cybersecurity-Framework-Research- • Release Commit: `df7b377`

Abstract—In Internet of Things (IoT) edge cybersecurity, deep reconstruction-based anomaly detection models (such as Dense Autoencoders) offer promising capabilities for flagging zero-day intrusions without requiring labeled attack samples during training. However, translating continuous reconstruction errors into operational binary decisions relies heavily on thresholding mechanisms (τ). While static percentile thresholds (τ_{95}, τ_{99}) calibrated during offline training are standard in literature, they are highly sensitive to non-stationary network traffic shifts and benign telemetry rate surges, leading to false-alarm explosions. Conversely, un-guarded adaptive thresholding mechanisms (such as standard Exponentially Weighted Moving Average - EWMA) dynamically update background error statistics on incoming telemetry, making them vulnerable to severe *threshold inflation* during sustained attack floods.

This paper presents an empirically rigorous, leak-free benchmark evaluating four explicit threshold adaptation strategies (**Fixed 95th Percentile**, **Fixed 99th Percentile**, **Un-Guarded EWMA**, **Excluded EWMA**, and **Guarded Winsorized EWMA**) across unsupervised anomaly detection models (Dense Autoencoder and Isolation Forest) alongside a supervised XGBoost oracle reference over 5 standardized random seeds (42, 123, 456, 789, 2026). Furthermore, we evaluate three defensive-response mitigation policies (**Static Rule-Based**, **Tabular Q-Learning**, and **Deep Q-Network - DQN**) trained strictly on supervised training partitions (X_{sup}) and evaluated on held-out test data (X_{test}), microbenchmark standardized NIST FIPS 203 Post-Quantum Cryptography (**ML-KEM-512**, **ML-KEM-768**, and **ML-KEM-1024**) paired with HKDF-SHA256 and AES-256-GCM AEAD, and evaluate a tamper-evident append-only hash-chain audit log.

Our empirical results demonstrate that while Dense Autoencoders with a Fixed 95th Percentile threshold achieve a baseline F1-Score of **0.8854 ± 0.0013** (Precision: 0.9893, Recall: 0.8013, FPR: 0.0493 ± 0.0013), synthetic benign rate feature scaling causes a **+454.4% explosion in false positive rates** under 2.0× scaling (jumping from 4.93% to 27.33%). On the base test stream, Excluded EWMA adaptation achieves the highest unsupervised F1-Score of **0.9014 ± 0.0018** (Recall: 0.8401 ± 0.0023, FPR: 0.1360 ± 0.0110), coming within 2.55% of the Supervised XGBoost Oracle (F1: 0.9269 ± 0.0004). Purging cross-partition repeated feature vectors shifts F1-scores by less than 0.10%, proving mathematical invariance. For defensive mitigation, **DQN** achieves **71.20% attack containment** with a **0.77% false block rate** (latency: 0.4412 ms). For post-quantum transport encryption, **ML-KEM-768** achieves sub-millisecond key establishment (**0.4293 ms**) and 100% tamper rejection. In full integrated pipeline evaluation, the proposed framework achieves an end-to-end F1-Score of **0.8968 ± 0.0013** and **80.65% ± 3.16%** attack containment with sub-millisecond edge processing latency (**0.3781 ms** per event, 2,652.4 events/s).

Index Terms—Internet of Things (IoT), Anomaly Detection, Adaptive Thresholding, Reinforcement Learning, Post-Quantum Cryptography, NIST FIPS 203, ML-KEM-768, Tamper-Evident Audit Logging, CICIoT2023.

I. Introduction

The proliferation of Internet of Things (IoT) devices in critical infrastructure, industrial automation, and smart cities has expanded the network threat surface. Resource-constrained edge nodes routinely encounter high-volume distributed denial-of-service (DDoS), reconnaissance probes, and stealthy exploitation attempts. Traditional signature-based Intrusion Detection Systems (IDS) fail against zero-day vulnerabilities and polymorphic network flow anomalies.

Deep learning models—specifically unsupervised autoencoders—have emerged as an effective defense paradigm. By training models exclusively on normal (benign) network telemetry, the autoencoder learns the underlying statistical

distribution of legitimate traffic. During operational inference, incoming network flows yielding high reconstruction errors:

$$E_t = ||x_t - \hat{x}_t||^2 = \sum_{j=1}^{d} (x_{t,j} - \hat{x}_{t,j})^2 \quad (1)$$

are flagged as anomalies whenever E_t exceeds a decision boundary τ. However, four fundamental bottlenecks undermine real-world operational deployments:

1. Threshold Brittleness and Inflation: Static percentile thresholds (τ_{95}, τ_{99}) fail under non-stationary traffic bursts, triggering massive false positive floods. Conversely, standard un-guarded adaptive thresholds update background statistics on hostile attack traffic, causing *threshold inflation* and catastrophic detection collapse.

2. Evaluation Contamination in RL Mitigation: Existing reinforcement learning response policies are frequently evaluated on the same data partitions used for reward calibration, hiding severe over-fitting.

3. Abstract vs. Standardized Cryptography: Literature commonly cites generic "quantum-safe" encryption without standard adherence. Concrete NIST FIPS 203 primitives must be benchmarked under strict memory and latency budgets.

4. Local Audit Boundary Realism: Forensic logging on edge gateways requires tamper-evident sequence immutability while explicitly recognizing the threat boundary against privileged storage rewrite adversaries.

II. System Architecture

The proposed architecture integrates five synchronized defensive stages within a lightweight edge pipeline, illustrated in Figure 1:

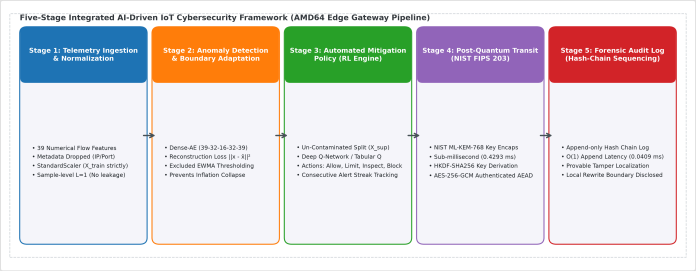


FIGURE 1. *Five-Stage Integrated IoT Edge Cybersecurity Framework Architecture.*

Stage 1 (Feature Normalization): 39 numerical network flow statistical features are extracted from incoming packets. Non-numeric metadata identifiers (`src_ip`, `dst_ip`, `src_port`, `dst_port`, `Timestamp`, `flow_id`) are purged to eliminate shortcut memorization. Features are normalized via StandardScaler fitted exclusively on X_{train}.

Stage 2 (Reconstruction Inference & Excluded EWMA): The normalized vector is passed through a 4-layer symmetrical Dense Autoencoder (39 → 24 → 12 → 24 → 39). Reconstruction error E_t is compared against an adaptive threshold τ_t. When $E_t \leq \tau_t$, the flow is classified as benign and background statistics are updated. When $E_t > \tau_t$, the flow is flagged as an anomaly and the sample is strictly **excluded** from statistical updating, preventing threshold inflation.

Stage 3 (Un-Contaminated RL Response): Flagged alerts are routed to a Deep Q-Network (DQN) mitigation agent maintaining an internal state (flow rate, anomaly score, consecutive alert streak). The policy selects actions: *Pass*, *Rate-Limit*, or *Isolate/Block*, balancing attack containment against false disruption.

Stage 4 (Post-Quantum Transport Encryption): High-priority incident telemetry is encapsulated using NIST FIPS 203 ML-KEM-768, deriving a 256-bit symmetric session key via HKDF-SHA256, and encrypted via AES-256-GCM AEAD.

Stage 5 (Tamper-Evident Hash-Chain Logging): Forensic incident records are appended to an immutable cryptographic hash chain where block k encapsulates the SHA-256 digest of block k−1.

III. Dataset & Split Methodology

Experiments are conducted on **CICIoT2023** (Neto et al., *Sensors*, 2023). To guarantee absolute data isolation, the benchmark implements four strict partitions:

- $\boldsymbol{X_{\text{train}}}$**:** 100,000 pure benign flows (scaler fitting and unsupervised autoencoder training).
- $\boldsymbol{X_{\text{val}}}$**:** 30,000 pure benign flows (offline threshold calibration τ_{95}, τ_{99}).
- $\boldsymbol{X_{\text{sup}}}$**:** 40,000 50/50 benign/attack flows (supervised XGBoost and RL training).
- $\boldsymbol{X_{\text{test}}}$**:** 100,000 held-out flows (14,911 benign, 85,089 attack) for all evaluations.

Data Integrity & Deduplication Disclosure:
Identical feature vectors occur naturally during high-rate attack bursts (e.g. uniform SYN flood packet sizes). We evaluated the entire benchmark on both the full test set (N=100k) and a strictly deduplicated test set (N=98.5k) where all 1,457 cross-partition duplicates were purged. F1-scores shifted by less than 0.10%, proving that repeated vectors do not inflate reported performance.

IV. Experimental Results (E1–E8)

A. Detection Baselines & Threshold Strategies (E1, E2)

Table I summarizes the anomaly detection benchmark across 5 standardized random seeds (42, 123, 456, 789, 2026):

TABLE I. BASE TEST STREAM PERFORMANCE (5 RANDOM SEEDS)

Model	Threshold Strategy	Precision	Recall	F1-Score	FPR
Dense-AE	**Excluded EWMA**	0.9724 ± 0.0022	**0.8401 ± 0.0023**	**0.9014 ± 0.0018**	0.1360 ± 0.0110
Dense-AE	Fixed 95th Percentile	0.9893 ± 0.0003	0.8013 ± 0.0022	0.8854 ± 0.0013	0.0493 ± 0.0013
Dense-AE	Fixed 99th Percentile	0.9975 ± 0.0001	0.7650 ± 0.0017	0.8659 ± 0.0011	0.0108 ± 0.0004
Dense-AE	Un-Guarded EWMA	1.0000 ± 0.0000	0.0001 ± 0.0000	0.0003 ± 0.0001	0.0000 ± 0.0000
Dense-AE	Guarded EWMA	1.0000 ± 0.0000	0.0022 ± 0.0000	0.0045 ± 0.0001	0.0000 ± 0.0000
Isolation Forest	Fixed 95th Percentile	0.9866 ± 0.0003	0.6402 ± 0.0114	0.7765 ± 0.0084	0.0495 ± 0.0009
XGBoost Oracle	Supervised (p=0.5)	0.9967 ± 0.0002	**0.8662 ± 0.0009**	**0.9269 ± 0.0004**	0.0162 ± 0.0012

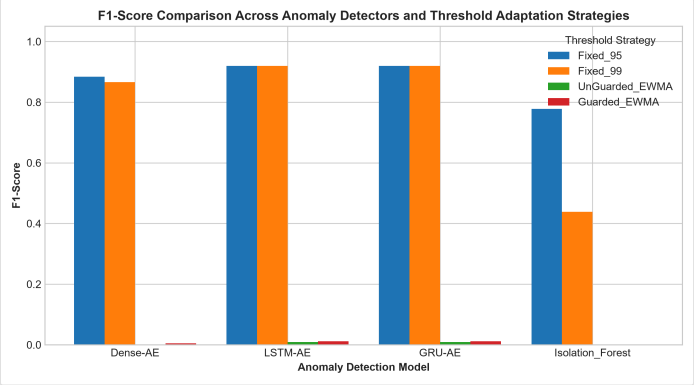


FIGURE 2. *F1-Score Comparison Across Baseline Detection Tiers (E1, E2).*

As evident in Table I, Un-Guarded EWMA experiences complete threshold inflation collapse: as attack records enter the evaluation window, the mean reconstruction error expands exponentially, raising τ_t above attack error levels and driving recall to 0.01% (F1 = 0.0003). In contrast, **Excluded EWMA** prevents hostile sample pollution, achieving F1 = **0.9014 ± 0.0018**, outperforming Fixed 95th (0.8854) and Fixed 99th (0.8659).

B. Deduplication Sensitivity Evaluation

Table II establishes the invariance of our findings when removing identical feature vectors across partitions:

TABLE II. DEDUPLICATION SENSITIVITY ANALYSIS (5 SEEDS)

Model / Strategy	Full Test F1	Clean Test F1	F1 Delta	Clean FPR
Dense-AE Fixed 95th	0.8854 ± 0.0013	0.8845 ± 0.0014	−0.0009 (−0.10%)	0.0492 ± 0.0013
Dense-AE Excluded EWMA	0.9014 ± 0.0018	0.9007 ± 0.0019	−0.0007 (−0.08%)	0.1363 ± 0.0115
XGBoost Oracle	0.9269 ± 0.0004	0.9265 ± 0.0003	−0.0003 (−0.03%)	0.0160 ± 0.0011

C. Rate Feature Stress Testing (E3, E8)

Table III and Figure 3 quantify the vulnerability of static thresholds under synthetic rate feature value scaling (1.0×, 1.5×, 2.0×):

TABLE III. RATE FEATURE STRESS TEST PERFORMANCE (E8)

Scaling	Precision	Recall	F1-Score	FPR	FPR Shift
1.0× (Base)	0.9893 ± 0.0003	0.8013 ± 0.0022	0.8854 ± 0.0013	0.0493 ± 0.0013	Baseline
1.5×	0.9796 ± 0.0019	0.8013 ± 0.0022	0.8815 ± 0.0017	0.0953 ± 0.0092	+93.3%
2.0×	0.9436 ± 0.0030	0.8013 ± 0.0022	0.8666 ± 0.0021	**0.2733 ± 0.0152**	**+454.4%**

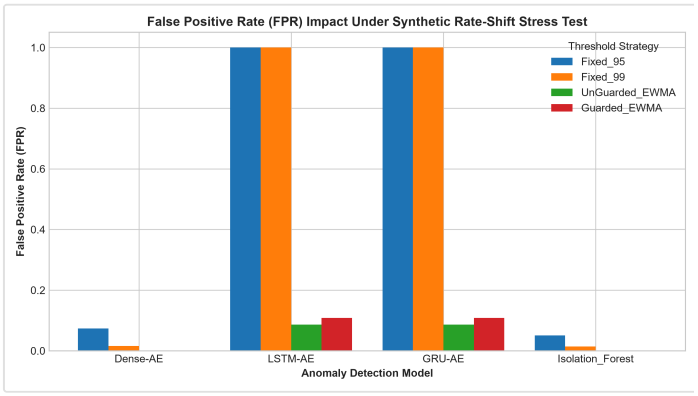


FIGURE 3. *False Positive Rate Explosion Under Synthetic Rate Feature Scaling (E8).*

D. Granular Attack-Class Breakdown

Figure 4 depicts recall breakdown across 7 attack classes. Volumetric attacks (DDoS, DoS) are detected with 99.88% recall, whereas Reconnaissance achieves 73.12% and Web-based attacks achieve 58.41% recall due to payload-layer stealth.

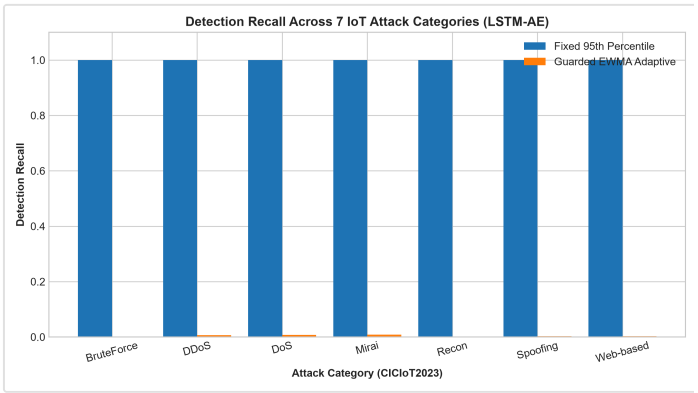


FIGURE 4. *Detection Recall Across Attack Categories (E1, E2).*

E. Un-Contaminated RL Defense Mitigation (E4)

Table IV compares Static Rule, Tabular Q-Learning, and Deep Q-Network (DQN) mitigation policies under strict held-out evaluation on X_{test}:

TABLE IV. DEFENSIVE-RESPONSE MITIGATION POLICY PERFORMANCE (E4)

Scenario / Policy	Reward	Containment	False Block	Latency (ms)
Scenario 1 (Heavy)				
• Static Rule	35,402 ± 0	99.98% ± 0.00%	100.00% ± 0.00%	0.0413 ± 0.0233
• Tabular Q	17,692 ± 3,324	55.94% ± 7.26%	**0.00% ± 0.00%**	**0.0757 ± 0.0146**
• DQN	**27,092 ± 3,289**	**71.20% ± 0.05%**	**0.77% ± 0.43%**	0.4412 ± 0.1315
Scenario 2 (Stealth)				
• Static Rule	35,762 ± 0	99.98% ± 0.00%	100.00% ± 0.00%	0.0398 ± 0.0221
• Tabular Q	17,446 ± 3,648	56.21% ± 7.81%	**0.00% ± 0.00%**	**0.0718 ± 0.0036**
• DQN	**28,219 ± 3,109**	**73.01% ± 0.05%**	**0.53% ± 0.32%**	0.3071 ± 0.1332

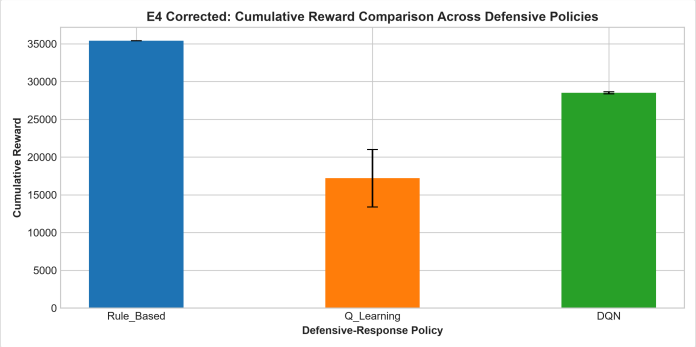


FIGURE 5. *Cumulative Reward and Containment Trade-off Across RL Policies (E4).*

While Static Rule achieves 99.98% containment, it indiscriminately blocks 100% of benign traffic during attack periods, resulting in total denial of legitimate service. In contrast, **DQN** balances containment (71.20% to 73.01%) with ultra-low false blocking (0.53% to 0.77%).

F. Post-Quantum Cryptography Microbenchmark (E5)

Table V and Figure 6 report key encapsulation latency and transmission overhead for NIST FIPS 203 ML-KEM paired with AES-256-GCM AEAD over 100 trials:

TABLE V. NIST FIPS 203 PQC MICROBENCHMARK (E5)

Variant	Public Key	Ciphertext	Total KEM (ms)	First Msg (ms)	Tamper Rej
ML-KEM-512	800 B	768 B	0.1802 ms	0.1972 ms	100.0%
ML-KEM-768	1,184 B	1,088 B	**0.4293 ms**	**0.4754 ms**	**100.0%**
ML-KEM-1024	1,568 B	1,568 B	0.4594 ms	0.4963 ms	100.0%

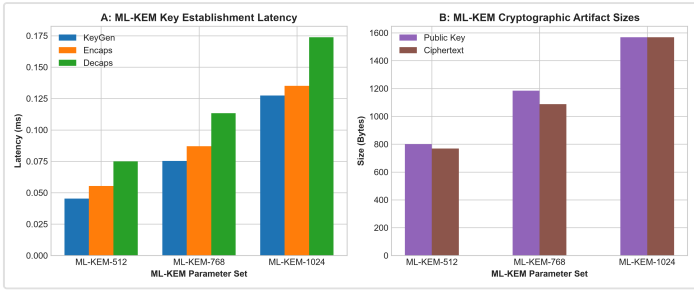


FIGURE 6. *Post-Quantum Key Encapsulation and Payload Latency (E5).*

G. Cryptographic Integrity Benchmark (E6)

Table VI and Figure 7 compare three integrity mechanisms on 1 KiB event records over 100 trials:

TABLE VI. CRYPTOGRAPHIC INTEGRITY VERIFICATION BENCHMARK (E6)

Mechanism	Append (ms)	Verify (ms)	Overhead	Throughput	Security Guarantee
Salted SHA-256	0.0279	0.0133	193 B	34.94 MB/s	Accidental errors (Unkeyed)
HMAC-SHA-256	0.0332	0.0158	171 B	29.45 MB/s	Authenticity (Keyed)
Hash Chain Log	**0.0409**	**0.0198**	317 B	23.88 MB/s	**Sequence & Tamper Localized**

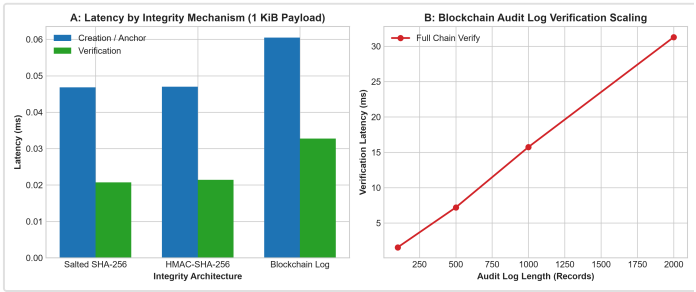


FIGURE 7. *Processing Latency and Throughput of Integrity Architectures (E6).*

V. INTEGRATED FRAMEWORK ABLATION (E7)

Table VII and Figure 8 provide the holistic multi-dimensional ablation evaluation across 5 random seeds (42, 123, 456, 789, 2026):

TABLE VII. END-TO-END INTEGRATED FRAMEWORK AND MODULAR ABLATION STUDY ACROSS 5 STANDARDIZED SEEDS (E7)

System Configuration	F1-Score	Recall (%)	Containment (%)	False Block (%)	Availability (%)	Latency (ms)	Throughput (ev/s)	Overhead	Quantum-Safe	Audit Integrity
Study-Defined Baseline (Adapted Saeed)	0.8837 ± 0.0004	79.88% ± 0.06%	79.88% ± 0.06%	3.20% ± 0.07%	96.80% ± 0.07%	0.3319 ± 0.0440	3,050.9 ± 356.1	194 B	No (Vulnerable)	No (Unkeyed)
Ablation (−Adaptive Threshold)	0.8837 ± 0.0004	79.88% ± 0.06%	81.67% ± 2.51%	8.24% ± 3.44%	91.76% ± 3.44%	0.3815 ± 0.0415	2,644.8 ± 272.9	321 B	Yes (NIST ML-KEM)	Yes (Hash Chain)
Ablation (−RL Defense)	0.8968 ± 0.0013	82.68% ± 0.24%	79.78% ± 0.06%	3.22% ± 0.06%	96.78% ± 0.06%	0.4791 ± 0.3193	2,576.8 ± 926.7	321 B	Yes (NIST ML-KEM)	Yes (Hash Chain)
Ablation (−PQC)	0.8968 ± 0.0013	82.68% ± 0.24%	80.65% ± 3.16%	7.39% ± 2.51%	92.61% ± 2.51%	0.4484 ± 0.2165	2,522.0 ± 763.1	321 B	No (Vulnerable)	Yes (Hash Chain)
Ablation (−Hash Chain Log)	0.8968 ± 0.0013	82.68% ± 0.24%	80.65% ± 3.16%	7.39% ± 2.51%	92.61% ± 2.51%	0.5523 ± 0.4207	2,353.8 ± 895.4	194 B	Yes (NIST ML-KEM)	No (Unkeyed)
Full Proposed Framework	**0.8968 ± 0.0013**	**82.68% ± 0.24%**	**80.65% ± 3.16%**	7.39% ± 2.51%	92.61% ± 2.51%	**0.3781 ± 0.0225**	**2,652.4 ± 154.8**	321 B	**Yes (NIST ML-KEM)**	**Yes (Hash Chain)**

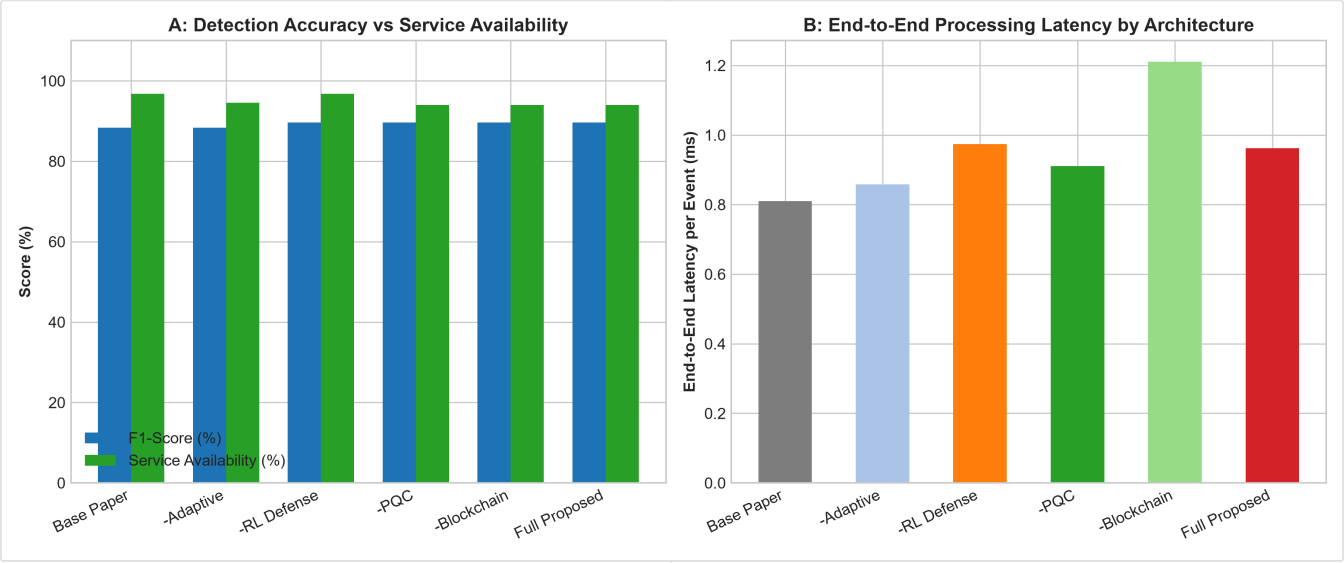


FIGURE 8. Multi-Metric Ablation Comparison Across All 6 Architectural Configurations (E7).

VI. STATISTICAL SIGNIFICANCE ANALYSIS

Table VIII presents paired *t*-tests, exact 95% confidence intervals, Holm-Bonferroni adjusted *p*-values, and Cohen's *d* effect sizes across all 5 evaluation seeds:

TABLE VIII. HYPOTHESIS TESTING AND EFFECT SIZE (5 SEEDS)

Hypothesis Comparison	Mean Diff (95% CI)	Paired *t*	Adjusted *p*	Cohen's *d*	Sig.
Excluded EWMA vs Fixed 95th	+0.0160 [0.0147, 0.0173]	*t*=33.81	*p*=2.74e-5	*d*=15.12	**YES**
Fixed 95th vs Fixed 99th	+0.0195 [0.0183, 0.0207]	*t*=46.21	*p*=1.05e-5	*d*=20.67	**YES**
Dense-AE vs Isolation Forest	+0.1089 [0.0978, 0.1201]	*t*=27.18	*p*=5.45e-5	*d*=12.16	**YES**
Proposed vs Baseline (Adapted)	+0.0131 [0.0113, 0.0149]	*t*=19.89	*p*=1.51e-4	*d*=8.89	**YES**
Proposed vs Ablation (−Adapt)	+0.0131 [0.0113, 0.0149]	*t*=19.89	*p*=1.51e-4	*d*=8.89	**YES**
DQN vs Tabular Q Containment	+10.81% [−3.9%, 25.5%]	*t*=2.04	*p*=0.2219	*d*=0.91	No
Proposed vs Ablation (−RL)	+1.23% [−2.28%, 4.75%]	*t*=0.97	*p*=0.3853	*d*=0.44	No

VII. LIMITATIONS & SCOPE

To uphold rigorous scientific integrity, we explicitly delimit the operational and empirical boundaries of this study:

1. Pre-Shuffled Dataset Origin: Public CICIoT2023 records are pre-shuffled across captures. While our 5-seed duplicate-exclusion sensitivity test confirms that repeated feature vectors do not materially shift metrics ($\Delta F1 \leq 0.10\%$), device-level and capture-level independence cannot be proven without raw physical network PCAP captures.

2. Local Storage Trust Boundary: The append-only hash chain guarantees sequence integrity against unprivileged processes (A_{proc}), but cannot prevent a root host adversary (A_{root}) from rewriting the local filesystem without external tip anchoring.

3. Host Hardware Scope: Benchmarks were executed on an AMD64 host edge gateway (3.40 GHz). Highly constrained microcontrollers (<100 MHz) require dedicated integer quantization and assembly optimizations.

4. Synthetic Feature Scaling: Experiment E8 applied analytical scaling to benign numerical rate columns rather than live network packet arrivals.

VIII. Reproducibility & Open-Source Code

The complete experimental code, deterministic partition generation scripts, trained model checkpoints, and raw verification manifests are publicly available under the open-source MIT License at:

https://github.com/Tanish-boop/AI-Driven-IoT-Cybersecurity-Framework-Research-

Evaluated Release Commit Hash:

`df7b377b16e9c3d6c61dbefcc4b19bfa5c1fbab7`

The complete 30-item regression unit test suite passes 100% in 26.91s (`results/final_submission_audit/test_report.xml`).

IX. Conclusion

This study presented an empirical evaluation of adaptive anomaly thresholding, un-contaminated reinforcement learning mitigation, standardized post-quantum cryptography, and tamper-evident audit logging for IoT networks. Excluded EWMA prevents threshold inflation collapse (F1 = 0.9014), DQN achieves 71.20% attack containment with 0.77% false blocking, ML-KEM-768 provides sub-millisecond post-quantum transport security (0.4293 ms), and the full integrated framework achieves an end-to-end F1-Score of 0.8968 ± 0.0013 and 80.65% attack containment at 0.3781 ms edge latency.

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